

List Comprehensions

COMS 261

1. Evaluate each of the following list comprehensions.

(a) `[x + 5 for x in [0, 2, 4, 6]]`

(b) `[word.upper() for word in ["cat", "dog", "frog"]]`

(c) `[x * 10 for x in range(6) if x > 2]`

(d) `[(2 * x) % 7 for x in range(7)]`

2. Write a list comprehension to generate this list:

`[1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024]`

3. Write a list comprehension with f-strings to generate the following list of strings.

`["4 * 1 = 4", "4 * 2 = 8", "4 * 3 = 12", "4 * 4 = 16", "4 * 5 = 20"]`

4. Use a list comprehension to re-write the following code using only one line.

```
1 | result = []  
2 | for n in range(1, 20, 2):  
3 |     result.append(1 / n)
```

5. Use a list comprehension to re-write the following code using only one line.

```
1 | result = []  
2 | for word in words:  
3 |     if len(word) > 5:  
4 |         result.append(word.upper())
```

6. The Indian mathematician Madhava of Sangamagrama (1340 - 1425) discovered a simple formula for the number π :

$$\frac{4}{1} + \frac{-4}{3} + \frac{4}{5} + \frac{-4}{7} + \dots + (-1)^k \frac{4}{2k+1} + \dots$$

- (a) Write a list comprehension to generate a list with the numbers $(-1)^k \frac{4}{2k+1}$ for $0 \leq k < 1,000$.

- (b) Use the Python **sum** function to calculate the sum of the list above. Is your total close to $\pi = 3.14159$?